

Lecture 4 Simple Linear Regression II

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Materials in this lecture notes come from two lecture notes written by Jean Kyung Kim (Inha Univ.) and Pratheepa Jeganathan (Stanford Univ.).



Outline

- ▶ Inference on simple linear regression model
- ▶ Prediction
- ▶ Example

Inference for β_0 or β_1

- ▶ Recall our model

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i,$$

where errors ε_i are independent $N(0, \sigma^2)$.

- ▶ In our heights example, we might want to know if there really is a linear association between **Daughter** = Y and **Mother** = X .
 - ▶ This can be answered with a *hypothesis test* of the null hypothesis $H_0 : \beta_1 = 0$.
 - ▶ This assumes the model above is correct, but that $\beta_1 = 0$.
- ▶ Alternatively, we might want to have a range of values that we can be fairly certain β_1 lies within.
 - ▶ This is a *confidence interval* for β_1 .

Setup for inference

- ▶ We can show that

$$\widehat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{\sum_{i=1}^n (X_i - \bar{X})^2}\right).$$

- ▶ Therefore,

$$\frac{\widehat{\beta}_1 - \beta_1}{\sigma \sqrt{\frac{1}{\sum_{i=1}^n (X_i - \bar{X})^2}}} \sim N(0, 1).$$

- ▶ The other quantity we need is the *estimator of standard error (SE)* of $\widehat{\beta}_1$.

- ▶ This is obtained from estimating the variance of $\widehat{\beta}_1$, which, in this case means simply plugging in our estimate of σ , yielding

$$\widehat{SE}(\widehat{\beta}_1) = \widehat{\sigma} \sqrt{\frac{1}{\sum_{i=1}^n (X_i - \bar{X})^2}} \quad \text{independent of } \widehat{\beta}_1$$

Testing $H_0 : \beta_1 = \beta_1^0$

$$Z \sim N(0,1), V \sim \chi^2(k) \left. \begin{array}{l} Z \text{ and } V \text{ are indep.} \end{array} \right\} \rightarrow T = \frac{Z/\sqrt{1/k}}{\sqrt{V/k}} \sim t(k), \quad \hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$$

$$\frac{(n-2)\hat{\sigma}^2}{\sigma^2} \sim \chi^2(n-2), \quad \hat{\beta}_1 \text{ and } \hat{\sigma}^2 \text{ are indep.}$$

- Suppose we want to test that β_1 is some pre-specified value, β_1^0 (this is often 0: i.e. is there a linear association)

- Under $H_0 : \beta_1 = \beta_1^0$, $H_1 : \beta_1 \neq \beta_1^0$

$$T = \frac{\hat{\beta}_1 - \beta_1^0}{\hat{\sigma} \sqrt{\frac{1}{\sum_{i=1}^n (X_i - \bar{X})^2}}} = \frac{\hat{\beta}_1 - \beta_1^0}{\frac{\hat{\sigma}}{\sigma} \cdot \sigma \sqrt{\frac{1}{\sum_{i=1}^n (X_i - \bar{X})^2}}} \sim t(n-2).$$

$$\frac{Z}{\sqrt{\frac{(n-2)\hat{\sigma}^2}{\sigma^2} / (n-2)}} \sim t(n-2)$$

$$Z = \frac{\hat{\beta}_1 - \beta_1^0}{\sigma \sqrt{1/S_{xx}}}$$

- Reject $H_0 : \beta_1 = \beta_1^0$ if $|T| \geq t_{\alpha/2}(n-2)$.



Example

Wage example

- ▶ Let's perform this test for the wage data.

```
SE.beta.1.hat = (sigma.hat * sqrt(1 /  
  sum((wages$education - mean(wages$education))^2)))  
Tstat = (beta.1.hat - 0) / SE.beta.1.hat  
data.frame(beta.1.hat, SE.beta.1.hat, Tstat)
```

```
##    beta.1.hat SE.beta.1.hat    Tstat  
## 1 0.07859951    0.004262471 18.43989
```

Wage example

- Let's look at the output of the `lm` function again.

Call:

```
lm(formula = logwage ~ education, data = wages)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-1.78239	-0.25265	0.01636	0.27965	1.61101

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.239194	0.054974	22.54	<2e-16 ***
education	0.078600	0.004262	18.44	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.4038 on 2176 degrees of freedom

Multiple R-squared: 0.1351, Adjusted R-squared: 0.1347

F-statistic: 340 on 1 and 2176 DF, p-value: < 2.2e-16

Wage example

- ▶ We see that R performs this test in the second row of the `Coefficients` table.
- ▶ It is clear that wages are correlated with education.

Why reject for large $|T|$?

- ▶ Observing a large $|T|$ is unlikely if $\beta_1 = \beta_1^0$: reasonable to conclude that H_0 is false.
- ▶ Common to report p -value:

$$\mathbb{P}(|T_{n-2}| \geq |T_{obs}|) = 2\mathbb{P}(T_{n-2} \geq |T_{obs}|)$$

```
2*(1 - pt(Tstat, wages.lm$df.resid))
```

```
## [1] 0
```

Confidence interval based on Student's t distribution

- ▶ Suppose we have a parameter estimate $\widehat{\theta} \sim N(\theta, \sigma_\theta^2)$, and the estimator of standard error $\widehat{SE}(\widehat{\theta})$ such that

$$\frac{\widehat{\theta} - \theta}{\widehat{SE}(\widehat{\theta})} \sim t(\nu).$$

- ▶ We can find a $(1 - \alpha) \cdot 100\%$ confidence interval by:

$$\widehat{\theta} \pm \widehat{SE}(\widehat{\theta}) \cdot t_{\alpha/2}(\nu).$$

- ▶ To prove this, expand the absolute value as we did for the one-sample CI

$$1 - \alpha \leq \mathbb{P}_\theta \left(\left| \frac{\widehat{\theta} - \theta}{\widehat{SE}(\widehat{\theta})} \right| < t_{\alpha/2}(\nu) \right).$$

Confidence interval for regression parameters

- ▶ Applying the above to the parameter β_1 yields a confidence interval of the form

$$\hat{\beta}_1 \pm \widehat{SE}(\hat{\beta}_1) \cdot t_{\alpha/2}(n-2).$$

- ▶ We will need to compute $\widehat{SE}(\hat{\beta}_1)$. This can be computed using this formula

$$\widehat{SE}(a_0\hat{\beta}_0 + a_1\hat{\beta}_1) = \hat{\sigma} \sqrt{\frac{a_0^2}{n} + \frac{(a_0\bar{X} - a_1)^2}{\sum_{i=1}^n (X_i - \bar{X})^2}}$$

with $(a_0, a_1) = (0, 1)$.

$$\textcircled{1} E(a_0 \hat{\beta}_0 + a_1 \hat{\beta}_1) = a_0 \beta_0 + a_1 \beta_1 \rightarrow \begin{matrix} a_0=0, a_1=1 \rightarrow \beta_1 \\ a_0=1, a_1=0 \rightarrow \beta_0 \end{matrix}$$

$$\begin{aligned} \textcircled{2} \text{Var}(a_0 \hat{\beta}_0 + a_1 \hat{\beta}_1) &= \text{Var}(a_0 \hat{\beta}_0) + \text{Var}(a_1 \hat{\beta}_1) + 2 \text{Cov}(a_0 \hat{\beta}_0, a_1 \hat{\beta}_1) \\ &= \text{Var}(a_0 (\bar{Y} - \hat{\beta}_1 \bar{x}) + a_1 \hat{\beta}_1) \\ &= \text{Var}(a_0 \bar{Y} + (a_1 - a_0 \bar{x}) \hat{\beta}_1) \\ &= \text{Var}(a_0 \bar{Y}) + \text{Var}((-a_0 \bar{x} + a_1) \hat{\beta}_1) + 2 \text{Cov}(a_0 \bar{Y}, (-a_0 \bar{x} + a_1) \hat{\beta}_1) \\ &= a_0^2 \text{Var}(\bar{Y}) + (a_1 - a_0 \bar{x})^2 \text{Var}(\hat{\beta}_1) \\ &= \frac{1}{n} a_0^2 \cdot \sigma^2 + (a_1 - a_0 \bar{x})^2 \frac{\sigma^2}{S_{xx}} \\ &= \left(\frac{a_0^2}{n} + \frac{(a_1 - a_0 \bar{x})^2}{S_{xx}} \right) \sigma^2 \end{aligned}$$

$$\text{Cov}(\bar{Y}, \hat{\beta}_1) = 0$$

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{x}$$

$$\begin{aligned} \textcircled{2} \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) &= \text{Cov}(\bar{Y} - \hat{\beta}_1 \bar{x}, \hat{\beta}_1) = \text{Cov}(\bar{Y}, \hat{\beta}_1) - \bar{x} \text{Cov}(\hat{\beta}_1, \hat{\beta}_1) \\ &= 0 - \bar{x} \cdot \frac{\sigma^2}{S_{xx}} = - \frac{\bar{x} \sigma^2}{S_{xx}} \end{aligned}$$

Confidence interval for regression parameters

- ▶ We also need to find the quantity $t_{\alpha/2}(n - 2)$. This is defined by

$$\mathbb{P}(T_{n-2} \geq t_{\alpha/2}(n - 2)) = \alpha/2.$$

- In *R*, this is computed by the function `qt`.

```
alpha = 0.05  
n = nrow(wages); n
```

```
## [1] 2178
```

```
qt(1-0.5*alpha, n-2)
```

```
## [1] 1.961055
```

- ▶ Not surprisingly, this is close to that of the normal distribution, which is a Student's t with ∞ for degrees of freedom.

```
qnorm(1 - 0.5*alpha)
```

```
## [1] 1.959964
```

- ▶ We will not need to use these explicit formulae all the time, as R has some built in functions to compute confidence intervals.

```
L = beta.1.hat -
  qt(0.975, wages.lm$df.resid) * SE.beta.1.hat
U = beta.1.hat +
  qt(0.975, wages.lm$df.resid) * SE.beta.1.hat
data.frame(L, U)
```

```
##              L              U
## 1 0.07024057 0.08695845
```

```
confint(wages.lm)
```

```
##              2.5 %      97.5 %
## (Intercept) 1.13138690 1.34700175
## education   0.07024057 0.08695845
```

Predictions

The estimation of the mean response

- ▶ Given $Y = \beta_0 + \beta_1 x + \epsilon$ and the least squares estimators of β_0 and β_1 are $\hat{\beta}_0$ and $\hat{\beta}_1$, respectively.
- ▶ For a chosen value x_0 , what is the prediction value of the **mean response variable**?
 - ▶ We need to estimate $\mathbb{E}[Y|x_0] = \beta_0 + \beta_1 x_0$.
 - ▶ Let $\mathbb{E}[Y|x_0] = \mu_0$ so $\mu_0 = \beta_0 + \beta_1 x_0$.
 - ▶ The best estimator for μ_0 is $\hat{\mu}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$.
- ▶ $\mathbb{V}[\hat{\mu}_0] = \mathbb{V}[\hat{\beta}_0 + \hat{\beta}_1 x_0]$.
- ▶ $\widehat{\text{SE}}(\hat{\mu}_0) = \hat{\sigma} \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}, \hat{\sigma}^2 = \frac{SSE}{n-2}$.
 - ▶ The estimation is much more accurate around $\bar{x}$.
- ▶ $\hat{\mu}_0 \sim N(\mu_0, \mathbb{V}[\hat{\mu}_0])$.

Predicting the response of an individual observation

$$E(\hat{Y}_0 | x_0) = \hat{\beta}_0 + \hat{\beta}_1 x_0, \quad \hat{Y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$$

$(Y_1, x_1) \dots (Y_n, x_n)$

$\hat{\beta}_0, \hat{\beta}_1$

$Y_0 = \beta_0 + \beta_1 x_0 + \epsilon_0$

ϵ_0 at $\frac{\epsilon_0}{\sigma}$

- ▶ Given $Y = \beta_0 + \beta_1 x + \epsilon$ and the least squares estimators of β_0 and β_1 are $\hat{\beta}_0$ and $\hat{\beta}_1$, respectively.
- ▶ For a chosen value x_0 , what is the prediction value of the response variable Y_0 ? Here Y_0 is a random variable.

▶ $Y_0 \sim N(\mathbb{E}[Y|x_0], \sigma^2)$.

▶ We took $\mathbb{E}[Y|x_0] = \mu_0$.

▶ The best estimator for Y_0 is $\hat{\mu}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0 = \text{Var}(Y_0) + \text{Var}(\hat{\beta}_0 + \hat{\beta}_1 x_0)$

$\text{Var}(Y_0 - \hat{\mu}_0) = \text{Var}(Y_0 - \hat{\beta}_0 - \hat{\beta}_1 x_0)$
 $= \text{Var}(Y_0 - (\beta_0 + \beta_1 x_0))$

- ▶ The predicted response distribution is the predicted distribution of the residuals $Y_0 - \hat{\mu}_0$ at the given point x_0 . So the variance is given by $\mathbb{V}[Y_0 - \hat{\mu}_0] = \mathbb{V}[Y_0] + \mathbb{V}[\hat{\mu}_0]$

▶ $\widehat{SE}(Y_0 - \hat{\mu}_0) = \hat{\sigma} \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}}$

Comparing SE of predicted response and mean response

- ▶ $\widehat{SE}(Y_0 - \hat{\mu}_0) > SE(\hat{\mu}_0)$.
 - ▶ Greater uncertainty in predicting one observation than in estimating the mean response.
 - ▶ Averaging in the mean response reduces the variability.

Confidence interval for mean response

- We can show that

$$\frac{\hat{\mu}_0 - \mu_0}{\widehat{SE}(\hat{\mu}_0)} \sim t(n-2).$$

- $(1 - \alpha)$ 100% confidence interval for μ_0 is

$$\hat{\mu}_0 \pm t_{\alpha/2}(n-2)\widehat{SE}(\hat{\mu}_0).$$

- Confidence limits.

$$\hat{\beta}_0 + \hat{\beta}_1 x_0 \pm t_{\alpha/2}(n-2) \sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}} \cdot s^2 = \frac{1}{n-2} SSE$$

Prediction interval

- ▶ We can show that

$$\frac{Y_0 - \hat{\mu}_0}{\widehat{SE}(Y_0 - \hat{\mu}_0)} \sim t(n-2).$$

- ▶ $(1 - \alpha)$ 100% prediction interval for Y_0 is

$$\hat{Y}_0 \pm t_{\alpha/2}(n-2) \widehat{SE}(Y_0 - \hat{\mu}_0).$$

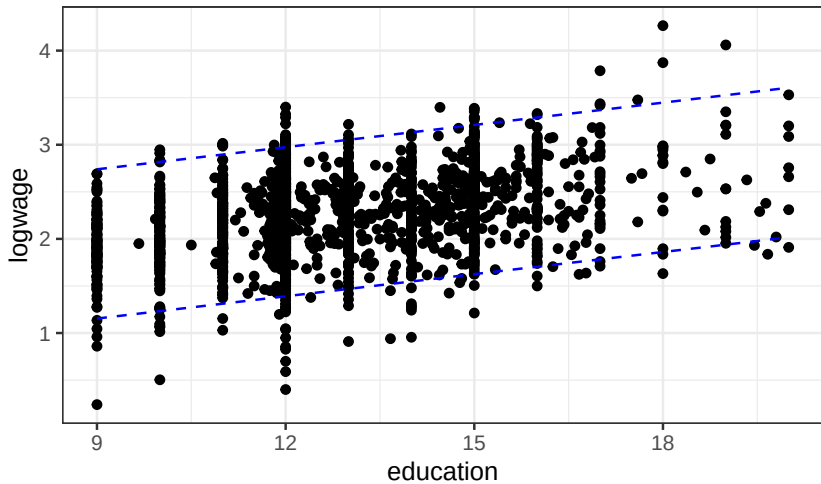
- ▶ Prediction limits.

$$\hat{\beta}_0 + \hat{\beta}_1 x_0 \pm t_{\alpha/2}(n-2) S \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}, \quad S^2 = \frac{1}{n-2} SSE$$

Wages vs. education example

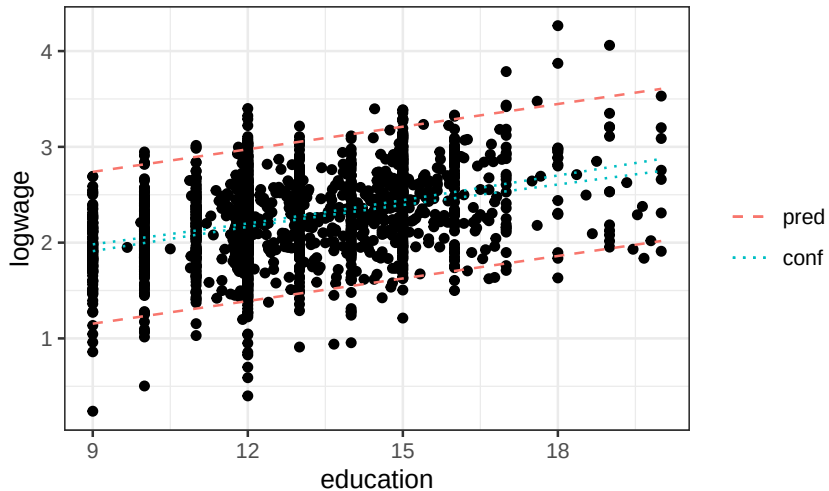
- Construct CI for the mean response for a sequence of x .

```
fname = './wage.csv'
wages = read.table(fname, sep=',',
                    header=TRUE)
wages.lm = lm(logwage ~ education,
               data = wages)
xval = data.frame(education = seq(min(wages$education),
                                  max(wages$education), length.out = 100))
prediction_bands = predict(wages.lm, xval,
                           interval = "prediction")
```



- Construct prediction intervals for the response for a sequence of x .

```
xval = data.frame(education =  
  seq(min(wages$education),  
      max(wages$education), length.out = 100))  
confidence_bands = predict(wages.lm, xval,  
  interval = "confidence")
```



References for this lecture

- ▶ Based on the lecture notes of [Pratheepa Jeganathan](#)
- ▶ Based on the lecture notes of [Jonathan Taylor](#) .